1. Title Page

Lab Name: Lab 13: Wireless LANs (WLAN) - WiFi and 802.11 Analysis **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

Wireless Local Area Networks (WLANs) operate under the **IEEE 802.11 (WiFi)** standard. Key concepts include:

1. **IEEE 802.11 Frame Format**: Unlike Ethernet frames which use a 2-address format (Source and Destination), 802.11 frames use up to a **4-address format**. This is because frames are transmitted over a shared wireless medium to an intermediate **Access Point (AP)**, which acts as a bridge to the wired distribution system.
2. **Addressing Mapping**: The roles of the address fields (Address 1, Address 2, Address 3, Address 4) are determined by the `To DS` (To Distribution System) and `From DS` flags in the Frame Control field:
   - `To DS = 0, From DS = 1` (Downlink from AP to Client): Address 1 is Destination (Client), Address 2 is BSSID (AP), Address 3 is Source (Sender).
   - `To DS = 1, From DS = 0` (Uplink from Client to AP): Address 1 is BSSID (AP), Address 2 is Source (Client), Address 3 is Destination (Receiver).
3. **802.11 Frame Types**: Frames are classified into:
   - **Management Frames**: Beacons, probe requests/responses, association requests/responses. (Beacons are broadcast periodically by the AP to advertise the network SSID, rates, and capabilities).
   - **Control Frames**: RTS/CTS (Request/Clear to Send) to avoid collisions and solve the hidden terminal problem, and ACKs to confirm frame delivery.
   - **Data Frames**: Encapsulate the actual network layer payloads.
4. **WPA2 4-Way Handshake**: Wireless links are encrypted to prevent eavesdropping. When a client associates, it performs a **4-way cryptographic handshake** using EAPOL (EAP over LAN) key exchanges to verify the Pre-Shared Key (PSK) and generate temporal encryption keys (PTK and GTK).

In this lab, you will deploy a wireless network in CML using a WiFi Access Point and two WiFi Client nodes, perform packet captures on the wireless interface, and analyze management, control, data, and security handshake sequences.

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3. Objective / Purpose

To configure a basic WLAN infrastructure in CML, capture and analyze 802.11 management frames (beacons), inspect 802.11 frame address fields and control flag maps, and analyze the WPA2 4-way cryptographic key exchange in Wireshark.

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4. Topology Diagram

```
graph TD
    H0["Host H0 (WiFi Client)<br>IP: 192.168.1.10/24<br>SSID: CS457-WiFi"] -.-> |"802.11 Link"|
WAP0["WiFi AP WAP0<br>SSID: CS457-WiFi<br>BSSID: Radio MAC"]
    H1["Host H1 (WiFi Client)<br>IP: 192.168.1.11/24<br>SSID: CS457-WiFi"] -.-> |"802.11 Link"|
WAP0

    classDef client fill:#d4f1f9,stroke:#007b8b,stroke-width:2px;
    classDef ap fill:#fff2cc,stroke:#d6b656,stroke-width:2px;
```

```
class H0,H1 client;
class WAP0 ap;
```

Details:

- 1x WiFi Access Point (`<initials>-WAP0`) broadcasting SSID `CS457-WiFi` .
- 2x WiFi Clients (`<initials>-H0` and `<initials>-H1`) associated with the AP.

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 2x WiFi Client Nodes, 1x WiFi Access Point Node

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6. Configuration / Methodology

WLAN Configuration Table

Device	CML Node Name	Interface	IP Address	SSID	Security Mode	Pre-Shared Key
Access Point	`<initials>-WAP0`	Radio0	192.168.1.1/24	`CS457-WiFi`	WPA2-PSK (AES)	`wirelesspass123`
Client 0	`<initials>-H0`	wlan0	192.168.1.10/24	`CS457-WiFi`	WPA2-PSK (AES)	`wirelesspass123`
Client 1	`<initials>-H1`	wlan0	192.168.1.11/24	`CS457-WiFi`	WPA2-PSK (AES)	`wirelesspass123`

Step-by-Step Configuration Guide

1. Access Point Setup (`<initials>-WAP0`)

Open the **Edit Config** or configuration pane of the WiFi AP node in CML:

1. Set the hostname to `<initials>-WAP0` (e.g., `KH-WAP0`).
2. Set the wireless SSID to `CS457-WiFi` .
3. Set the encryption type to WPA2-Personal (PSK) with AES.
4. Configure the Pre-Shared Key (password) to `wirelesspass123` .
5. Assign a static IP address to the AP's local interface: `192.168.1.1/24` .

2. Client Host Setup (`<initials>-H0` and `<initials>-H1`)

On each client host node, open the **Edit Config** tab to connect to the wireless network:

1. Configure Host `<initials>-H0` hostname and wireless association parameters:

   ```
   # Configure wlan0 interface
   ip addr add 192.168.1.10/24 dev wlan0
   ```

```
# Associate with the CML WiFi AP
iwconfig wlan0 essid CS457-WiFi key s:wirelesspass123
```

2. Configure Host `<initials>-H1` hostname and wireless association parameters:

```
ip addr add 192.168.1.11/24 dev wlan0
iwconfig wlan0 essid CS457-WiFi key s:wirelesspass123
```

Step-by-Step Traffic Generation & Execution:

1. **Start Packet Capture**: Right-click the wireless radio link or AP node interface, select **Packet Capture**, and click **Start**.
2. **Verify Client Association**: Open Host H0's console and verify connection status:

```
iwconfig wlan0
```

(Confirm that the Access Point MAC address is displayed, indicating successful association).

3. **Generate Traffic**: Ping Client H1 from Client H0:

```
ping -c 4 192.168.1.11
```

4. **Trigger 4-Way Security Handshake**: To capture the initial key negotiation, disconnect and reconnect Host H1 while capture is running:

```
# On Client H1: Disconnect and reconnect
ifconfig wlan0 down
sleep 2
ifconfig wlan0 up
iwconfig wlan0 essid CS457-WiFi key s:wirelesspass123
```

5. **Download Capture**: Stop the CML packet capture and download the `.pcap` capture file to your local computer.

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7. Wireshark Analysis and Questions

Open your downloaded packet capture in Wireshark and answer the following questions:

Part A: 802.11 Management Frames (Beacons)

1. Locate a **Beacon frame** sent by the AP (`<initials>-WAP0`).
   - What is the Source MAC address (the transmitter)?
   - What is the BSSID of the wireless network? Does this match the AP's MAC?
2. Inspect the **Beacon parameters** in the packet details pane:
   - What SSID is advertised?
   - What WPA/WPA2 security requirements (e.g. Robust Secure Network - RSN group/pairwise cipher suites) are listed in the parameters?

Part B: 802.11 Frame Addressing & Control Flags

3. Locate an **802.11 Data frame** containing your ping traffic (ICMP request) sent from Client H0 to Client H1.
    - Expand the **Frame Control** field in the IEEE 802.11 header. What are the binary/hex values of the `To DS` and `From DS` flags?
    - Based on these flag states, why are these values set (what is the directional path of the packet)?
4. Inspect the address fields in the header of this data frame:
    - What is the MAC address value for:
        1. Address 1 (Receiver Address / RA)
        2. Address 2 (Transmitter Address / TA)
        3. Address 3 (Source/Destination Address depending on DS flags)
    - Map these addresses to the actual CML nodes in your topology (H0, H1, or WAP0).
5. Explain why an IEEE 802.11 wireless frame header requires up to **four** MAC address fields, whereas a standard Ethernet II wired frame header only requires **two**. (Discuss the bridging role of the AP).

Part C: WPA2 Security and the 4-Way Handshake

6. Locate the EAPOL key negotiation packets during Host H1's reconnection.
    - How many EAPOL packets are exchanged to complete the **4-Way Handshake**?
    - What type of keys (PTK, GTK, PMK) are dynamically generated during this handshake?
    - Why is the 4-Way Handshake necessary (what does it protect against in a wireless environment)?

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Summarize your findings. Contrast the operations of 802.11 wireless networks with standard 802.3 wired Ethernet, detailing the role of management frames in network advertising, 3/4-address mapping over bridged access points, and the necessity of EAPOL handshakes for link-layer security.

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10. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- A single PDF report containing:
    - Your written answers to the 6 Wireshark analysis questions in Section 7.
    - A screenshot of your active CML topology showing the running hosts (`<initials>-H0` , `<initials>-H1`) and AP (`<initials>-WAP0`).
    - A screenshot of Host H0's console showing the output of `iwconfig wlan0` , confirming association.
    - An annotated screenshot of your Wireshark capture window showing the WPA2 4-Way Handshake EAPOL key exchange.